Quadraloop: real-time audio looping software

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Abstract

This paper will discuss the efficiency of a software designed specifically for the creation and generation of variable audio loops. Thus, while proposing a new computer tool for musical expression linked to a specific compositional process (looping), I wanted to explore and personalise a compositional language that would integrate a new form of expression in the professional music world. The development of the software Quadraloop gave me the opportunity to explore a new way to create music. Rather than using tools yielded by existing musical traditions, I created a link with these traditions through technical research and programming. By giving emphasis on a specific parameter, it was easier to develop a unique graphical environment to explore new musical possibilities.

1 Introduction

Composition procedures today are widely dominated by the presence of computers. The specialised field of computer music represents the outcome of the accumulated composition techniques of the twentieth century. This is true in the instrumental domain as well as in the electroacoustic music domain. The great musical inventions are now represented virtually and it is now possible to simulate various sound emissions and precise composition techniques in very elaborate ways. Henceforth, the act of composing combines technical innovation and musical expression. In the past years, new writing techniques were rarer, which gave the opportunity to merge various musical languages. A reflection or a fortaste of the globalising era? "A new way to make music rather than creating a new musical style"[1].

2 Quadraloop

The software Quadraloop was developed to adhere to the following criteria: to explore and exploit the variability of an audio loop in a real- time situation, in the context of electroacoustic composition. The application generates the necessary sonic material for the realisation of an entire musical work. The most important aesthetic criteria is repetition of course. This follows an already existent musical style called Minimalism. But the goal of this software is not to recreate the Minimalist musical style. There are several musical genres, which promote repetition as a vehicle of expression. This is why Quadraloop focuses on repetition as a phenomena and its different aspects. This software can therefore be used in various situations.

3 Musical applications

The three main functions of Quadraloop:

1- To compose a piece consisting of audio loops

2- To generate new sonic material in real-time

3- To create a loop for later usage

3.1 Composing

The basic construction of Quadraloop is done within Max/MSP's graphical programming environment (MacOS). The composer in the computer age is confronted with a new way to create. Not only does the composer need to be musically creative, but also needs to be technically inclined to successfully express new musical ideas. At least, this is the case in electroacoustic music. Did the act of composing shift toward programming and technical innovation? Maybe not to such an extent, however, it is at least necessary to be fluent in some technical or programming language as they are now indissociable with artistic creativity.

Quadraloop is designed to be used in several musical applications. The main application is the construction of an entire piece based on looping and repetition. It is possible to display a timeline with the representation of all the parameters affecting the imported or pre-recorded sound files. The visual score is then executed and it is possible to record the result. This then becomes a composition, in the sense of traditional electroacoustic (acousmatic) music.

3.2 Real-time Performance

An area in constant expansion is real-time performance and real-time manipulation of sound. Quadraloop emphasises this specific current in music and favours interaction between artist and technology. This important aspect in the conceptualisation of a virtual instrument enhances the role of the musician, in broadcasting artistic expression. Traditional electroacoustic music favours individual work in the studio. Nevertheless, "the possibility of *double immediate* widens the perspective of the artists, composers and musicians. Expand the instrument and project it in space,

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reinvent musical writings, widening the field of musical interpretation, such are the possibilities offered by these new technical and musical means"[2]. It is very interesting to use Quadraloop as an instrument generating sound instantaneously, without necessarily having a direct control over the final result. This leaves room for interpretation and adds an almost tangible human dimension to the performance.

3.3 Sound Design

The Quadraloop application can be used as an accessory to another music creation environment, as a tool for sound design. It is possible to create short musical instances, either melodic or rhythmic, for later insertion in other musical works. Because of the design of the interface, it is also easy to create new sounds otherwise impossible to simulate in more conventional graphic environments. For example, the variable indexing within a file permits a variable sweeping using both *loop start* and *loop end* indices. The following is a symbolic demonstration of what is called variable looping. The 26 letters of the alphabet represent the audio file, and the letters in boldface print designate the loop. Here, it is easy to observe the behaviour of the *loop start* and *loop end* indices:

State 1 : A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

State 2 : **A B C D E F G H I J K L M** N O P Q R S T U V W X Y Z

State 3 : A B C D E **F G H I J K L M** N O P Q R S T U V W X Y Z

State 4 : A B C D E F G H I J **K L M N O P Q R** S T U V W X Y Z

State 5 : A B C D E F G H I J K L M N O **P Q** R S T U V W X Y Z

State 6: A B C D E F G H I J K L **M N** O P Q R S T U V W X Y Z

4 Interface

The design of the graphical user interface is of great importance for efficient use in a musical application. This phase is determinant in moulding the aesthetic outcome of any virtual instrument. Visual organisation of the different parameters is a critical element to facilitate maximum usage of a process. The position and assigned priority of each parameter control gives a certain emphasis on specific manipulations. For example, in Quadraloop, there are three main parameters for changing sound. They are represented by three rotary knobs (X Y and P), index, length and pitch (see Figure 1).

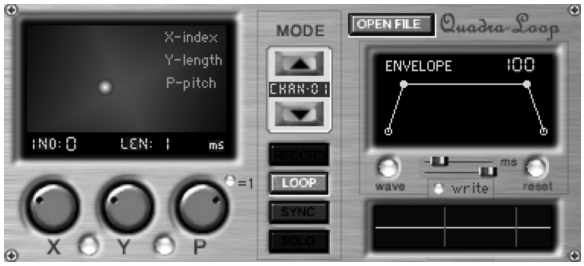


Figure 1. One of four looping modules in Quadraloop

The interface offers several graphical methods for processing sound, depending on the context and degree of desired control. A simple parameter of a loop can generate various musical results. In a two dimensional plane, for example, it is possible to control two parameters with a single click of a mouse. In the future, it would be nice to explore a three-dimensional space and have access to the three main parameters at once. This would certainly be possible with sensors, graphic tablets and external gestural controls.

5 Conclusion

As a composer, this project allowed me to develop new skills in an expanding field. I realised it was possible to invent your own creative musical instruments and share a personal artistic expression through formalisation of an abstraction. A structured approach may harvest a proper evolution of a distinctive musical idea. The framework of a technological process allows artists to enrich their vocabulary of expression and helps them find a new sense of musical direction. The potential of musical creation and artistic innovation is largely dependant on available tools and this has a growing influence on current technological developments. Quadraloop software works in this direction and enhances interaction with musicians. This novel graphical user interface for musical expression certainly blends well in the latest musical trends.

6 Demonstration

The possibilities with the Quadraloop software will be demonstrated with a real-time performance for violin (Ivan Zavada) and computer (Quadraloop software).

References

[1] Attali, J. 1989. *Bruits*. Fayard, Presse Universitaire de France, p. 33.

[2] Lévy, F. *Le temps réel en musique – CD-R*. IRCAM Publishing 2002.